



Supporting mentors working with students with intellectual disabilities in higher education

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Abstract

Project Panther LIFE is an inclusive postsecondary transition program for students with intellectual disabilities providing university access and participation with the primary goal of employment at program completion. Students in the program receive support from their academic mentors and peer coaches during the academic year. This study examines the skills and activities mentors use during their weekly sessions with students with intellectual disabilities and identifies areas in which mentors may require further support or training. Data analysis revealed major themes related to inclusion, self-determination, and adaptive behavior skills. Upon review of the data, we suggest that mentors need ongoing support from transition programs especially in areas related to encouraging self-advocacy and supporting time management.

Keywords

higher education, mentoring students, students with intellectual disabilities, transition program, student support

Introduction

The American job market has become more knowledge based, increasing the need for a post-secondary education (PSE) (Carnevale and Desrochers, 2003). Federal legislation such as the

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Higher Education Opportunity Act (HEOA, 2008) has made it possible for students with intellectual disabilities (IDs) to enroll in university programs through postsecondary transition models. The increased need for knowledge in the everyday workplace, in addition to newly acquired opportunities, draws students with ID to college campuses. With this increase of students with ID in PSE, it is necessary to examine the types of supports needed to ensure successful outcomes, both academically and socially. Peer mentorship has been identified as a crucial aspect of support and a practical approach for academic and social dilemmas (Campbell-Whatley, 2001). The literature regarding mentorship practices has been limited to mostly theoretical research. Nevertheless, one of the few studies conducted in a postsecondary transition program was Jones and Goble's (2012). Although it provided a thorough examination of mentorship practices, limitations of the study included a small research sample and a homogeneous group of mentors. Additionally, Jones and Goble (2012) stated that further in-depth analysis was needed regarding what skills mentors expended time on with their mentees (e.g. academic development and social skills development) as well as the activities performed to develop these skills (e.g. using vocabulary flash cards and meeting new people). The purpose of this article is to determine what activities and skills mentors of students with ID primarily focus on during mentoring sessions as well as to identify areas in which mentors may require further support or training.

Existing literature on postsecondary programs

An individual with ID is a person with a cognitive impairment with specific limitations in the areas of intellectual and cognitive functioning and adaptive behavior (HEOA, 2008). The most recent reauthorization of the HEOA (2008) outlines a series of provisions that focus attention and federal resources on the postsecondary goals of these students as outlined on their individualized education plans. In addition to the HEOA (2008) provisions for this population of students, the Individuals with Disabilities Education Act (IDEA, 2004) defines the transition for students with disabilities as a coordinated set of activities designed within an outcome-oriented process, which promotes movement from school to postschool activities, including PSE, integrated employment, and independent living. The coordinated set of activities is guided by the student's preferences and interests using a person-centered model and includes instruction, community experiences, and the development of employment and other postschool adult living objectives and, when appropriate, acquisition of daily living skills and functional vocational evaluation.

According to Hart et al. (2006), students with ID have the poorest postschool outcomes compared to other students with disabilities especially when it comes to attending college and particularly participating in coursework. A comparison study conducted on 20 students with ID with some type of PSE experience (e.g. noncredit audit and certificate course) and 20 with no PSE experience revealed that students with PSE experience were much more likely to obtain competitive employment, required fewer supports, and earned higher wages. Furthermore, these same students had increased levels of self-esteem and expanded social networks (Zafft, Hart, & Zimbrich, 2004). The need for students with disabilities to become independent and economically self-sufficient is directly linked to attaining increased levels of knowledge and skills gained through access and participation in PSE. This need is the same for all individuals—those with and without disabilities—to be a part of a workforce and engaged members of society.

In the United States, there are approximately 242 PSE programs in 47 states and most are collaborative partnerships between institutions of higher education and the local public school system. Seventy-four percent of these programs support students who are dually enrolled in both

high school and college/university (18–22 years of age) while 33% are programs designed for adult students with ID beyond the age of 22 (Grigal et al., 2015). Thirty-six percent of families surveyed about their postschool plans indicate that their first choice is having their student with ID attend a 4-year college/university while 22% want a community college option (Grigal and Neubert, 2004). Additionally, very few PSE programs offer residential life experiences on campus. A survey conducted with 13 PSE programs in one state revealed that 87% of the 163 students participating in these programs were involved in employment training, 36% were enrolled in a typical college course, and over half participated in campus life activities. All exiting students were linked to an adult service agency or community rehabilitation program as they exited. Seventy-nine percent qualified for Social Security benefits, 84% had a job for the summer, and 65% exited with a paid job (Hart et al., 2001; Zafft et al., 2004).

Providing access to PSE for students with ID is only the beginning, while ensuring their active participation, full involvement, and completion is also critically important. While more students with disabilities enroll in 2-year community and technical colleges than in other type of postsecondary schools, currently there is no information available on their rate of PSE completion. But given the additional barriers (e.g. lack of acceptance and full accessibility/inclusion) individuals with disabilities often encounter in our society and within our learning systems, their opportunities may be limited and oftentimes in jeopardy unless the needed resources and supports are put in place.

Program description

Project Panther LIFE: Panther Learning Is For Everyone was conceptualized and guided by the regulations of the HEOA (2008) to support students with ID in the areas of academics, social activities, employment, and independent living and through an ongoing person-centered planning (PCP) process of exploration and discovery of interests and strengths. The PCP process allows for the formation of a circle of support for the student (Turnbull et al., 2011) by valuing and respecting the student's preferences for his or her future planning and establishing goals within a variety of domains (e.g. academic enrichment, career development and employment, and self-determination).

Florida International University (FIU) in Miami, Florida, has partnered with the local public school system and with a parent resource center to design, implement, and sustain a postsecondary transition program for students with ID in the Miami-Dade community. *Project Panther LIFE* is in its fourth year of implementation and its primary goal is to provide a comprehensive postsecondary program and system of supports (e.g. academic mentors) through a well-planned and individualized curriculum leading to paid employment at program completion.

This nondegree certificate program allows students with ID the flexibility to select and enroll in University courses in a variety of program areas (e.g. public speaking, computing, and music) and access and participate in University campus life. Besides access and participation, students also engage in a series of job shadowing experiences, complete community employment internships, and attend other required program activities (e.g. summer residential program). Refer to Table 1 for a list of program activities and descriptions.

Program design

Project Panther LIFE's curriculum framework was adapted from the 10 critical areas of independent living as proposed by Wehman and Kregel (1997) (e.g. self-determination, financial planning, etc.) and includes a detailed curriculum map which aligns with each transition domain

Table 1. Program activities and descriptions.

Program activity	Description
On-campus job shadowing experiences ($n = 3$)	These employability experiences occur on-campus with the support of the employability/community coordinator in collaboration with the student's assigned job coach and on-the-job supervisor. Each student will have a total of three job shadowing experiences to introduce him or her to a particular knowledge base and skill set (e.g. clerical/office, recreational, and child care). Job shadowing experiences are part-time, typically occurring two to three times per week.
Off-campus community employment internships ($n = 2$)	These employability internship experiences occur off-campus with the support of the employability/community coordinator in collaboration with the student's assigned job coach and on-the-job supervisor. Each student will have a total of two supervised internships with a public or private business in the local community to focus on developing expertise in an area of career interest. Internships are full-time, typically occurring five times per week.
Online functional academics curriculum	i-Ready Reading and Math Curriculum will be used by students in the program under the direction of the M-DCPS project coordinator. This curriculum focuses on the development of reading and math skills through online individualized instruction with ongoing progress monitoring.
Online transition curriculum	Unique Learning System Transition Program will be used by students in the program under the direction of the M-DCPS project coordinator. The primary goal of this curriculum is to increase a student's independence in the areas of job skills and daily living through the creation of learning opportunities in the context of real-world scenarios.
Online employability curriculum	MECA will be accessed by students in the program under the direction of the M-DCPS project coordinator. MECA is an age-appropriate transition, career exploration, career assessment, and vocational assessment system related to training, education, and employment.
Program activities	Program activities include attendance at University Orientation and Welcome Back meetings, weekly sessions with system of supports (e.g. academic mentors and peer coaches), attending monthly student meetings, family/student seminars, and so on.

MECA: Microcomputer Evaluation of Careers and Academics; M-DCPS: Miami-Dade County Public Schools.

provided by each member of the partnership (FIU, Miami-Dade County Public Schools, Parent to Parent of Miami, Vocational Rehabilitation). The domains of the curriculum framework are Employment, Internship, and Related Employment Activities, Academics and Technology, Socialization, Relationships, and Self-Esteem, Community Living, Self-Determination, Healthy and Safety, Personal Care and Hygiene, Home Living, Travel and Mobility, Financial Planning and Money Management, Electives and Memberships, and Continuing Education.

To ensure that students meet the domains set forth by the program, a program of study (POS) was also created that serves as a roadmap for the students in the *Panther LIFE* program. There are five major components in the POS including University courses, job shadowing experiences,

community internship experiences, online curricula, and required program activities (e.g. trainings and workshops and meeting with academic mentors and peer coaches).

An array of supports is provided to students to ensure successful completion of the program and potential for paid employability. Supports include academic advising and instructional programming from project coordinators and university faculty advisors; academic mentoring and peer coaching from FIU students; access to and participation in University activities and events, technology trainings, and family/student seminars. Each planned activity and support leads students to increased levels of knowledge and skills gained through access and participation in PSE.

Academic mentoring and social support are ongoing and provided by a group of trained FIU students who act as academic mentors and peer coaches. Mentors are provided with trainings and one-on-one consultations with a mentor coordinator throughout the academic year in order to better understand the needs of students with ID and how to implement a variety of instructional strategies. On a weekly basis, mentors are asked to schedule three sessions lasting 1 h each to work one-on-one with their assigned *Panther LIFE* student.

The role of an academic mentor is to assist *Panther LIFE* students in accessing university coursework while helping to foster student academic independence and ownership of learning. Job performance responsibilities for each academic mentor include (1) help to maintain a positive academic environment, (2) assist students with assignments in courses as assigned, (3) encourage appropriate classroom behavior, (4) foster student independence and encourage responsibility, (5) help students with time management, and (6) provide accommodations to facilitate students access to coursework. Typical mentoring activities have included re-explaining and/or clarifying content/information, proofreading assignments, and creating graphic organizers for studying key concepts.

The role of the peer coach varies only in that the focus of the work centers on assisting the *Panther LIFE* student in understanding and acclimating to University campus and social life. Job performance responsibilities for each peer coach include: (1) Help to understand and maintain a positive social environment; (2) Assist and encourage students with socially appropriate university behaviors; (3) Assist students with handling money in real life skills; and (4) Encourage independence and responsibility. Examples of peer coaching support have included introducing the mentee to others, addressing social skill needs, and attending on- and off-campus events. Providing this structured system of support ensures that students experience a high level of success during their program tenure.

Method

Description of setting

FIU is an urban, multicampus, public research university serving its students and the diverse population of South Florida. Since opening in 1972, FIU has achieved many benchmarks of excellence (e.g. 8th among international business programs in nation; 24th among the top universities in the country in terms of engagement and contributions to its community); has a nationally renowned faculty known for outstanding teaching and research of which 96% hold doctoral degrees or highest degree attainable in their fields; students from throughout the United States and more than 130 foreign countries (60% of the student body is Hispanic); 35% of teachers in Miami-Dade County Public Schools with an FIU degree; over 125,000 alumni who live and work in the South Florida area; and ranked 1st in the nation awarding bachelor's and master's degrees to Hispanic students. The University offers 180 bachelors, masters, and doctoral programs

in 12 colleges and schools. During the 2014–2015 academic year, FIU had 54,000 students enrolled, making it the largest university in South Florida and ranking it fifth on the list of largest public universities by enrollment.

Description of participants

Academic mentors and peer coaches (all referred to as “mentors” in this publication) consisted of 31 university students ages 18–42 of which 25 were females and 6 were males. University class standing of the group consisted of 11 seniors, 9 sophomores, 4 juniors, 1 freshman, and 6 graduate level students. The group’s racial/ethnic diversity consisted of 67% Hispanic (21), 14% White (4), 14% Black (4), and 5% Asian (2) students. The group had a variety of majors including special education (15%; 5), biology (20%; 6), hospitality and tourism management (12%; 4), nursing (10%; 3), psychology (5%; 2), and advertising, architecture, pre-medical studies, physical education, history, business administration, adult education and human resource development, and social work (5%; 1 respectively). Ninety percent (27) of mentors indicated that they had previous experience working with individuals with disabilities. Participants were selected through an application and interview process and worked with the students for at least one full academic semester. Seven students (22%) worked with the program for the entire duration of the study (Fall 2012 to Fall 2013).

Research design

Data collection

Panther LIFE acquired Institutional Review Board approval from FIU and all mentors agreed upon hire via consent form to complete an online questionnaire regarding their overall experiences and perceptions of the program and a log tracking activities and reflecting on their sessions with their assigned student. The mentors were trained during an initial orientation meeting on what type of information should be entered in the log, but no further probes were used during the process. Data from the logs and questionnaires were collected in December 2012, May 2013, August 2013, and again in early December 2013. In total, 58 logs and 55 questionnaires from 31 students were collected over the course of a 1-year period.

Instrumentation

The questionnaire is a summative evaluation tool compiled of 15 quantitative Likert-type scale questions and 3 qualitative open-ended response questions. For this particular study, only the qualitative data were used in the coding process. The open-ended questions consisted of general comments about the program, general comments about the mentees, and favorite or least favorite parts about working with the mentees.

The log is an open-ended data collection instrument that served as a journal and was used to collect formative data as students were instructed to fill out the log after each mentoring session. This log contained spaces to write date of session, time frame of session, activities accomplished, activities pending, and reflective thoughts. A new log was started at the beginning of each semester regardless of whether the mentor was continuing from the previous semester.

Table 2. Mentor reported themes and subthemes.

Theme	Subtheme	Frequently recorded activities
Inclusion	On-campus involvement	Homecoming/alumni activities Sporting events Social club activities Joining a club
	Off-campus involvement	Special olympics activities Going to the movies Attending festivals
Self-determination	–	Creating personal goals Utilizing the university disability Resource center Using public transportation Determining learning style
Adaptive behavior skills	Career development	University career services events Résumé/job interview preparation Observing student at work, sending e-mails, utilizing iPad for organization
	Social and communication skills	Meeting for lunch Practicing speechmaking Symposiums, presentations, and conferences
	Physical development/health and wellness	Going to the gym together Working on a diet and talking about nutrition Doing yoga
	Academic skills	Reading/answering book questions Doing homework and projects Studying together and going over study habits Meeting at and using the library

Data analysis

To assess the data and identify important aspects of the mentorship program along with potential areas of improvement, the researchers were guided by the coding process outlined by Gilbert (2008). Open coding was used after each set of logs and questionnaires were collected to determine emerging themes in each log. Axial coding was then used with all sets of open coding data to determine relationships in themes across the data sets. Emerging themes were then narrowed into subthemes and categories were developed.

Results

Three major themes were originally discovered in the data set: inclusion, self-determination, and adaptive behavior skills. Refer to Table 2 for major themes and subthemes. Within the inclusion category were found two subcategories of on-campus involvement and off-campus involvement. Within the adaptive behavior skills category were found five subcategories: career development, social and communication skills, physical development/health and wellness, and academic skills.

Inclusion

On-campus involvement. The majority of mentors stated a goal of having students become more involved on campus, especially in terms of joining clubs and groups. A few examples of mentors assisting students included intramural sports teams and political volunteer groups on campus. Mentors reported that joining a club on campus was beneficial during the university experience for social aspects. One mentor stated, "I want her to join a club that she can be actively involved on campus. Being involved in the club will help her open up." Another mentor indicated, "She needs to work on interacting with people and being less intimidated and more social. We will volunteer next week and this will be a great opportunity for her."

It was also reported that engaging in on-campus activities together helped to create rapport with the students as they immediately had something in common or a shared experience. One mentor reported that taking salsa dancing classes on campus with the student was her favorite part of the program. She reported, "It was a great bonding experience. She really enjoyed it and I loved dancing with my student each week!" Other mentors also agreed that having common interests outside of academics made for a more enjoyable experience.

On-campus involvement also included attending events such as social club events (i.e., sorority and fraternity events), football games, featured speakers, and holiday campus events. During these events, mentors shared that the students had the chance to meet new people and practice appropriate social behaviors. One mentor reported taking her assigned student to her club events to expand his social circle. She said, "This is a great way for him to make more friends. He participated without hesitation. He engaged appropriately and was nice to everyone he met." Other mentors were in agreement that on-campus events provided benefits to all parties involved.

Off-campus involvement. Along with on-campus involvement, mentors reported spending time together off-campus to make the relationship more authentic. One mentor indicated that when finding out the student held a common interest in a video game, the two began getting together off-campus with friends to play the game. He related, "Our online gaming has really been a nice way to spend quality time and socialize." Others recorded going to events (both program sponsored and nonprogram sponsored) off-campus such as movies, festivals, and fairs. Mentors indicated these off-campus events offered mentees a chance to practice social behaviors in places other than the University setting. One mentor also noted that finding activities off-campus incorporated technology into the sessions as the mentee would use an app on his i-Pad to find things to do in the area. A few mentors attended University conferences with their mentees and one commented, "[It was] such a fun time for everyone. We all bonded on a new level." Generally, mentors were in agreement that off-campus activities were beneficial, though some reported difficulty in planning events due to time constraints and transportation.

Self-determination

Mentors frequently mentioned the use of learning style quizzes disseminated at the initial orientation and training. Several mentors reported that filling out the learning style quiz helped them to support their mentees by understanding their strengths and challenges in the learning process. One mentor added that her mentee was able to open up and discuss his learning needs. Mentors shared that the quiz was a valuable tool to establish rapport with their mentees by learning about each other. As one stated, "Today we talked about what we thought our strengths and weaknesses were."

It ended up being a great ice breaker.” Other mentors also indicated that the learning style quiz led to setting semester goals.

Mentors reported several different ways in which self-determination was encouraged with their mentees including travel training, speaking to professors, and establishing an ongoing dialogue with their parents. Mentors mentioned assisting students with being on time for their transportation and planning ahead to ensure arrival in the correct place. One mentor went as far as to ride the public transportation with her mentee to get him acclimated. She reported, “I would really like to see him take advantage of the transportation resources provided to him. We will ride together so he feels more comfortable with the process.” Later in the semester the same mentor reported, “He really loves taking the bus now!” Yet another mentor reported helping his mentee research driving programs and opening a dialogue with his parents about learning how to drive. Other mentors reported going to class with their mentee to encourage them to speak to their professors about needed learning accommodations. Some mentors documented meeting with mentees’ parents and encouraging more levels of independence. One mentor shared, “Today we talked about some more leisure time that her parents could give her to have off campus activities possible. Her homework for the next meeting is to talk to them about this topic.” Engaging and incorporating families in mentoring has been recommended by McDonald et al. (2005) as one factor that may be effective in addressing poor adulthood outcomes for young adults with disabilities (e.g. assumed incapacity and lower levels of independence) and increasing the success of the mentoring work.

Adaptive behavior skills

A common theme among the data revolved around four adaptive behavior skills: career development, social and communication skills, physical development/health and wellness, and academic skills. It was also within this theme that many mentors reported challenges. Adaptive behavior skills were a focal point in the initial orientation and training, during which mentors were encouraged to model proper behaviors (e.g. communication skills) and use self-determination skills (e.g. problem-solving and decision-making).

Career development. Mentors indicated asking their mentees about their interests and future goals. A few mentors mentioned visiting their mentees at their job shadowing or paid job sites for observation. One mentor commented, “He got to show me what he does at work each day and his responsibilities.” Other mentors agreed that observation was a mutually benefitting experience. Additionally, mentors recorded activities to prepare mentees with the skills necessary for a workplace. Some activities included building a résumé, attending job fairs, conducting mock interviews, and building time management skills. One mentor said, “We sent his résumé to his advisor. I noticed he is developing a sense of professionalism.” Another mentor commented on working on her mentee’s time management skills: “I explained how it is rude to arrive late to anything; I emphasized the idea that if he was ever late to work they would fire him.” Mentors were in consensus that behavior should be congruent from mentor meetings to job sites and reinforced this throughout the study.

Technological skills were also emphasized in regards to adaptive behaviors for employment. All *Panther LIFE* students received an iPad and participated in trainings, and mentors reported assisting their mentees with utilizing this tool. The majority of mentors reported working with their mentees to send e-mail and use apps for organization, calendars, and learning. One mentor said, “He had a lot of difficulty with organization in general, and I tried to help by getting him an app

that will remind him about the things he needs to do.” Another reported, “She knows basic skills, but I am going to help her learn how to use Microsoft Word and PowerPoint.” Others reported using iPads for practicing functional academic skills such as reading and math. A mentor commented, “The games are fun and he really enjoys it while learning Math skills.” All mentors were in agreement that iPads are a useful tool for their mentee’s career development skills.

The data revealed that an area of challenge for mentors included the lack of time management and organizational skills exhibited by the students in the program which is a challenge often cited in the literature (Sowell and Maddox, 2015). Mentors referenced mentees missing appointments, being unorganized, not prioritizing items, and forgetting materials during classes or sessions. As one mentor commented at the end of the semester, “Overall, good experience to work with [her]; however, her organizational skills need to improve.” Another commented, “The biggest frustration was working on her time management skills.” Literature demonstrated that improved time management can lead to enhanced independence (Davies et al., 2002). This may be indicative of attention needed in relation to these skills by support staff and mentors in the future.

Social and communication skills. Social and communication skills play a major role in adaptive behavior and social competence, which directly affects future adjustment including employment (Gresham et al., 2001; Kupersmidt et al., 1990). Social behavior, communication, and interactions were emphasized in the initial orientation and training for the mentors, thereby drawing it to their attention during the study. Many mentors chose to focus on communication and appropriate social behavior with their mentees as a goal. Several mentors noted that seeing social growth and intellectual connections with their mentees was the most rewarding part of the experience.

A few common themes that occurred during the study regarding social and communication skills were appropriateness, public speaking, and expanding social circles. Nearly all mentors mentioned introducing their mentee to their circle of friends. One mentor added, “I introduced her to my friend who is majoring [in] Nursing; that’s the major that she wants to study. We had a conversation about that.” Several mentors spent weeks preparing and practicing speeches with their mentees who attended conferences with one mentor/mentee group making memorizing a speech a semester goal.

Although growth was reported, mentors expressed challenges with social behaviors and the need to redirect students to more appropriate behaviors. One mentor stated, “He tells so many girls they are beautiful; I explained how that might make them feel uncomfortable.” Continuing to train and support mentors and other support staff in implementing strategies to improve social and communication skills is a future consideration and is supported by the literature (Campbell-Whatley, 2001; Gresham et al., 2001; Kupersmidt et al., 1990).

Physical development/health and wellness. Teaching mentees about nutrition and workout plans was a common theme in the data collected. While the mentees attended sessions during the week at the Recreational Center, most mentors felt that additional help was needed. Activities reported in the logs included spinning and Zumba classes, weight training, and intramural sports. One mentor said, “I went to the gym with [him]; we came up with a workout plan that he should stick with and brought some workout exercises.” Other mentors reported progress with their workout goals for the semester. One mentor commented, “He called me the next day telling me he was sore but that he’s happy I made him do the extra two sets at the gym yesterday. I think he is seeing results!” Overall, mentors agreed that engaging in physical activities played an important role in the overall experience for their mentees.

Mentors also reported having conversations with their mentees regarding nutrition. One shared, "We talked about how he needs to be on a diet and too many French Fries are not good for him." Others reported their mentees used information from a nutrition class and applied it to their everyday life. One mentor reported a semester goal of helping her mentee make healthier choices at the store on campus. Overall, mentors reported an increase of healthy eating habits over the course of the academic year. Although nutrition was not discussed during the orientation and training meeting, healthy diets can reduce the risk of obesity, which occurs more frequently in people with intellectual disabilities (Marshall et al., 2003). Therefore, more consideration about proper nutrition and exercise may be necessary in the future.

Academic skills. Mentors were introduced to the stages of the learning process (e.g. acquiring, organizing, storing, and receiving information) and various instructional strategies for helping their mentees with academics, including note-taking tips, concept mapping, and the use of graphic organizers. Nearly all mentors reported working one-on-one with their mentee on academic work at some point during the study. Academic skills mentioned in the data included reading comprehension, vocabulary, math and calculation, study skills, researching in the library, and writing outlines for papers. One mentor recounted a success story with her student: "While preparing the outline for the essay, she had some trouble spelling out certain words. Rather than asking me, she went on her phone and used an app that allows her to spell check." A number of mentees were reported as verbally asking for help while others required prompts regarding assignments and due dates. Other mentors also shared that their mentees showed improvement in academics throughout the study. One mentor stated, "As the semester goes on she is taking responsibility for her own learning." Another mentor said, "I love how she is becoming an independent student." Study skills were also emphasized through note cards and reviewing notes and handouts. One mentor recounted her strategies for studying with her mentee: "She typed the notes by herself and offered to read information as I read and explain the information to her. Her homework was to take the notes home and highlight important details." Another mentor described giving vocabulary homework to her mentee to improve comprehension when reading for class. Overall, mentors reported attempting multiple study strategies with their mentees and ultimately witnessing positive and improved academic results at the end of each semester.

Alternatively, concerns from the mentors emerged when examining the data. Mentors reported feeling that online classes were difficult for their mentees due to the reliance on personal accountability. Several mentors also commented on the number of classes their mentees were taking per semester. "He was overwhelmed the entire semester and unable to keep up with 3 classes and a work schedule. It took away from his academics." The consensus was that mentoring sessions were not as effective when more than two classes per semester were selected. Other concerns included difficulty with assisting students with course assignments due to limited functional skills in reading and math.

Discussion

Federal legislation is now making it possible for students with ID to enroll in university programs through inclusive postsecondary transition models. With the increase in the number of students with ID participating in these programs, it is necessary to examine the types of supports that need to be in place for this population of students to experience success. One support that is often identified is the use of peer mentorship for improved academic and social development (Campbell-Whitley,

2001) and supporting the belief in the value of having a more experienced peer to provide guidance (Hill and Reddy, 2007).

In this study, we examined the mentoring support students received in a postsecondary transition program called *Panther LIFE*. The program's mentoring philosophy centers around establishing a relationship that is based on teaching and learning approach that is developmental in nature, views mentors as potential "agents of change," and assumes that the concept of mentoring is a valuable component of an effective postsecondary program (Sowell and Maddox, 2015).

The purpose of the study was to determine the specific activities and skills mentors primarily focused on during their sessions with students as well as to identify the areas in which mentors need further training to successfully support their students in postsecondary transition programs. The data reported from participating mentors revealed the activities and skills that were held in high priority during mentoring sessions focused on improving and enhancing the academic, social, and employability skills of the students. Mentors wanted their students to understand their academic responsibilities in university courses but equally stressed the importance of understanding and displaying appropriate social behavior on- and off-campus.

Furthermore, mentors understood that the goal of the program was to prepare students for paid employment and therefore engaged their students in conversations and activities related to career exploration and the development of work-related skills. These mentor perspectives allow transition programs such as *Panther LIFE* to reevaluate program priorities, focus on overall issues related to student access, inclusion, and future success, and as stated by Sowell and Maddox (2015), "provide a unique and potentially valuable insight into the influence of having students with IDs on a 4-year university campus" (p. 8). Thus, in this study, the data revealed the role of mentors as being multifaceted and extending beyond the "job description" to include mentors as role models, advocates, and friends as described by Colvin and Ashman (2010).

The results also demonstrate that mentors at the postsecondary level for students with ID oftentimes describe their activities as experiences that can be labeled as mutually beneficial to both parties as outlined in previous research (Eby et al., 2008; Lee et al., 2003) and thus sheds light on the mentoring partnership and process that is established. Furthermore, results revealed that mentors choose to focus on challenges that directly reflect their own as well as other university-aged students without ID such as becoming a self-advocate, attempting to be more independent, expanding social circles, gaining a better understanding of diversity as related to working with an individual with a disability, and gaining employment after graduation. Although students with ID may struggle with these challenges in different ways, ultimately mentors serve as a bridge of understanding between a university campus and a student. Researchers such as Jones and Goble (2012) have noted that mentors require preparation and training before working with students with ID; however, the results indicate that continued support and opportunities for learning throughout the transition program is needed. Mentors need further training and support from postsecondary transition programs in transferring their knowledge to their mentees, especially in areas related to the development of time management; balancing work life, school life, and social life, improving social skills; and assisting with academics and learning.

Upon review of the data and to use it for program improvement, *Panther LIFE* staff discussed and implemented new features to assist mentors. To offer further training and support, monthly meetings with mentors were arranged. These meetings ran 1 h in length and covered a variety of topics including balancing life and work, student motivation, and promoting a healthy lifestyle. Also, program staff revised the format of the mentor orientation and identified the need for a training handbook. Each mentor receives a training handbook outlining program goals, program

staff roles, contact lists, suggested strategies, and helpful resources (e.g. academic tips, graphic organizers). In addition, the format of the log was modified to be an online questionnaire that is filled out on a weekly basis to gather data on an ongoing basis instead of at the end of an academic semester. These modifications allow for program staff to be aware of potential areas of struggle, intervene with the appropriate level of support, and receive ongoing feedback from mentors throughout the semester.

Training and supporting mentors continues to be a struggle due to the nature of utilizing university students with time constraints and limited base knowledge regarding diverse populations. Additionally, addressing the variety of needs of students with ID in a time-limited training session poses a challenge to program staff to identify and cover the most critical topics (e.g. employment and instructional strategies) while not having sufficient time to cover other topics that may also be of importance (e.g. health and fitness). Future research may still be needed in the areas of assisting mentors with strategies for promoting appropriate social behaviors, accommodations for academic techniques, and reinforcing time-management skills.

In spite of these challenges, establishing well-planned and organized connections through mentoring partnerships focusing on student-centered goal attainment with traditional university students and students with ID in postsecondary inclusive programs needs to be the standard of practice as these relationships “add value” and enhance the lives of not just the students with disabilities but “enrich the educational experience of all students preparing them to more readily embrace differences as they enter the real world environment” (Sowell and Maddox, 2015: 9). The knowledge gained through understanding these unique relationships between mentor and mentee should then be used to inform our practices in designing and implementing effective and inclusive postsecondary programs for students with ID.

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References

- Campbell-Whately GD (2001) Mentoring students with mild disabilities: “The nuts and bolts” of program development. *Intervention in School and Clinic* 36(4): 211–216.
- Carnevale AP and Desrochers DM (2003) Preparing students for the knowledge economy: what school counselors need to know. *Professional School Counseling* 6(4): 228–236.
- Colvin JW and Ashman M (2010) Roles, risks, and benefits of peer mentoring relationships in higher education. *Mentor Tutor: Partnership Learn* 18(2): 121–134.
- Davies DK, Stock SE and Wehmeyer ML (2002) Enhancing independent time-management skills of individuals with mental retardation using a palmtop personal computer. *Journal Information* 40(5): 358–365.
- Eby LT, Allen TD, Evans S, et al. (2008) Does mentoring matter? A multidisciplinary meta-analysis comparing mentored and non-mentored individuals. *Journal of Vocational Behavior* 7(2): 254–267.
- Gilbert N (2008) *Researching Social Life*. 3rd ed. London: Sage.
- Gresham FM, Sugai G and Horner RH (2001) Interpreting outcomes of social skills training for students with high-incidence disabilities. *Exceptional Children* 67(3): 331–344.

- Grigal M, Hart D, Smith FA, et al. (2015) *Think College National Coordinating Center: Annual Report on the Transition and Postsecondary Programs for Students with Intellectual Disabilities (2013–2014)*. Boston: University of Massachusetts Boston, Institute for Community Inclusion.
- Grigal M and Neubert DA (2004) Parents' in-school values and post-school expectations for transition-aged youth with disabilities. *Career Development for Exceptional Individuals* 27(1): 65–85.
- Hart D, Grigal M, Sax C, et al. (2006, August) *Postsecondary Education Options for Students with Intellectual Disabilities*. Institute for Community Inclusion: Research to Practice, Issue 45.
- Hart D, Zafft C and Zimbrich K (2001) Creating access to college for all students. *The Journal for Vocational Special Needs Education* 23(2): 19–31.
- Higher Education Opportunity Act (HEOA) (2008) 20 U.S.C. §§ 1001–1161.
- Hill R and Reddy P (2007) Undergraduate peer mentoring: an investigation into processes, activities and outcomes. *Psychology Learning & Teaching* 6(2): 98–103.
- Jones MM and Goble Z (2012) Creating effective mentoring partnerships for students with intellectual disabilities on campus. *Journal of Policy and Practice in Intellectual Disabilities* 9(4): 270–278.
- Individuals with Disabilities Education Act (IDEA) of 1989, Pub. L. No. 101–476, 104 Stat. 1142 (1990).
- Kupersmidt JB, Coie JD and Dodge KA (1990) The role of poor peer relationships in the development of disorder. In: Asher SR and Coie JD (eds) *Peer Rejection in Childhood. Cambridge Studies in Social and Emotional Development*. New York: Cambridge University Press, pp. 274–305.
- Lee SH, Yoo SY and Bak SH (2003) Characteristics of friendships between children with and without mild disabilities. *Education and Training in Developmental Disabilities* 38(2): 157–166.
- Marshall D, McConkey R and Moore G (2003) Obesity in people with intellectual disabilities: the impact of nurse-led health screenings and health promotion activities. *Journal of Advanced Nursing* 41(2): 147–153.
- McDonald K, Balcazar FE and Keys CB (2005) Mentoring youth with disabilities. In: Dubois DL and Karcher MJ (eds) *Handbook of Youth Mentoring*. Thousand Oaks: Sage Publications, pp. 493–507.
- Sowell RL and Maddox B (2015) Added value: Perspectives of student mentors working within a university-level inclusion education program. *Online Journal of Education Research* 3(1): 1–10.
- Turnbull A, Turnbull R, Erwin EJ, et al. (2011). *Families, Professionals, and Exceptionality: Positive Outcomes Through Partnerships and Trust*. 6th ed. Upper Saddle River: Pearson.
- Wagner M, Newman L, Cameto R, et al. (2005) *After High School: A First Look at the Postschool Experiences of Youth with Disabilities*. Available at: http://www.nlts2.org/reports/2005_04/nlts2_report_2005_04_complete.pdf (accessed 12 March 2014).
- Wehman P and Kregel J (eds) (1997) *Functional Curriculum for Elementary, Middle, and Secondary Age Student with Special Needs*. Austin: Pro-Ed.
- Zafft C, Hart D and Zimbrich K (2004) College career connection: A study of youth with intellectual disabilities and the impact of postsecondary education. *Education and Training in Developmental Disabilities* 39(1): 45–53.